



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Using AI and Behavioral Finance to Cope with Limited Attention and Reduce Overdraft Fees

Daniel Ben-David, Ido Mintz, Orly Sade

To cite this article:

Daniel Ben-David, Ido Mintz, Orly Sade (2026) Using AI and Behavioral Finance to Cope with Limited Attention and Reduce Overdraft Fees. *Management Science* 72(1):204-222. <https://doi.org/10.1287/mnsc.2022.00304>

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License. You are free to download this work and share with others for any purpose, except commercially, and you must attribute this work as “*Management Science*. Copyright © 2025 The Author(s). <https://doi.org/10.1287/mnsc.2022.00304>, used under a Creative Commons Attribution License: <https://creativecommons.org/licenses/by-nc/4.0/>.”

Copyright © 2025 The Author(s)

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Using AI and Behavioral Finance to Cope with Limited Attention and Reduce Overdraft Fees

Daniel Ben-David,^a Ido Mintz,^b Orly Sade^{a,*}

^aFinance Department, Jerusalem School of Business, The Hebrew University of Jerusalem, Jerusalem 91905, Israel; ^bIntuit, Mountain View, California 94043

*Corresponding author

Contact: daniel.ben-david@mail.huji.ac.il (DB-D); ido_mintz@intuit.com (IM); orlysade@huji.ac.il,  <https://orcid.org/0000-0002-2651-1007> (OS)

Received: February 8, 2022

Revised: June 2, 2024; September 28, 2024

Accepted: October 6, 2024

Published Online in Articles in Advance:
May 14, 2025

<https://doi.org/10.1287/mnsc.2022.00304>

Copyright: © 2025 The Author(s)

Abstract. We test how effective a human–algorithm interaction is at stopping users from overdrawing their bank accounts. We use a randomized field experiment and draw our sample from users of a large personal financial management platform operating in the United States and Canada. We find that sending as-needed reminders is effective in and of itself, and the impact is intensified by the human response to the structure of the message. More simple messages are more effective, and the framing of the simplified message makes a difference. Users with medium to high annual incomes and users with fair to good credit scores are most likely to respond positively. We find that the investigated artificial intelligence solution reduces information-gathering costs and has a positive effect but is not sufficient in all cases. Those with challenging financial situations may find it harder to act upon the warning. For our analysis, we employ parametric identifications and time-to-event semiparametric analysis. Our work contributes to the literature on financial technology as advisors, human–computer interaction, limited attention, behavioral finance, and experimental finance.

History: Accepted by Jean-Edouard Colliard, Special Issue on the Human-Algorithm Connection.



Open Access Statement: This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License. You are free to download this work and share with others for any purpose, except commercially, and you must attribute this work as “Management Science. Copyright © 2025 The Author(s). <https://doi.org/10.1287/mnsc.2022.00304>, used under a Creative Commons Attribution License: <https://creativecommons.org/licenses/by-nc/4.0/>.”

Funding: O. Sade acknowledges financial support for this research from the Krueger Center and the Albertson-Waltuch Chair in Business Administration.

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mnsc.2022.00304>.

Keywords: human–computer interaction • artificial intelligence • behavioral finance • overdraft • limited attention • field experiment

1. Introduction

We use a randomized, large-scale field experiment to study the effectiveness of an artificial intelligence (AI)–based advising system. The system uses a human–algorithm interaction and is intended to reduce overdraft and nonsufficient fund (NSF) fees. The experiment utilizes an AI intervention system that predicts and evaluates the likelihood of overdrawing events using the financial information of customers from a large personal financial management application operating in the United States and Canada. The AI solution not only predicts overdrafts but also creates and sends recurring email messages as needed, acting as a personal guidance system regarding one’s future financial situation.

Individuals seek advisors’ guidance for many functions and decisions, and some businesses are exploring whether AI can fill some advising roles, for example, in

health and wellness, commerce, transportation, law, marketing and finance.¹ We investigate an AI innovation that tackles a widespread problem in household finance, which is overdrafts. The literature indicates that managing personal finances can be challenging: it demands substantial effort, time, literacy, and access to information. However, personal finance is important, and many households use several types of financial advisors to overcome their challenges (Mintel 2015). It follows that algorithms might be useful for guiding and mitigating some of these challenges.²

AI-based financial advising services have several potential advantages over human financial advisors. They can quickly process and analyze large amounts of information. They can provide more affordable guidance to those who cannot afford a personal advisor. They are accessible from many locations, which can

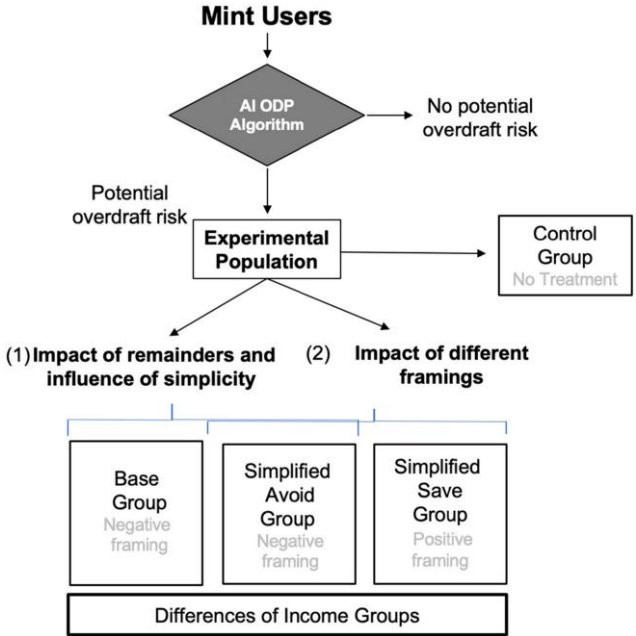
overcome geographical constraints. They can monitor a situation continuously and offer multiple interventions in a timely manner. Furthermore, they may be less prone to certain human biases. Hence, there is growing academic interest in evaluating these different AI advising mechanisms (e.g., Fisch et al. 2017, D’Acunto et al. 2019, among others).

Whereas AI can identify opportunities and suggest decision-making improvements, humans can still decide whether to act on them. We focus on the human decision to act or not when warned by an algo-based intervention that the person is at risk for incurring overdraft or NSF fees.³ An overdraft is a transaction that brings the account holder’s balance below zero.⁴ In the United States alone, retail banks charge \$17 billion annually as NSF fees,⁵ which accounts for approximately 6% of their total revenues (Stango and Zinman 2014) and for two thirds of the income they earn from fees.⁶ The economic literature offers several competing explanations for why people incur overdraft fees. The first is based on complete rational information modeling. Fully rational individuals may overdraw their accounts and pay the fees as a financing alternative after evaluating all the benefits of consumption and the associated overdraft costs. A second explanation relies on limited attention or rational inattention.⁷ Because acquiring information can be costly or complex, one might overdraw and incur the associated fees after considering only incomplete information. A third explanation relies on the ostrich effect and information-dependent utility models, in which individuals tend to avoid acquiring information at unpleasant times and incur costs for overdrawing their account.⁸

For our empirical investigation, we use Mint, a web-based personal financial management (PFM) application with more than 20 million users in the United States and Canada. Mint offers different services, such as personal financial insights and budget planning, to its customers. Mint users connect their bank accounts and credit cards to the application, which has a personalized interface. We focus on a Mint service named overdraft prediction (ODP), an algorithm that predicts an individual’s likelihood of running a below-zero balance. The ODP is a gradient-boosting decision tree model with periodic retraining that utilizes multiple financial traits.⁹ Our sample is Mint customers who were identified by the algorithm as having a high likelihood of paying an overdrawing fee in the immediate future. If the AI system identifies the user as exceeding a certain threshold on a prediction score, an email is sent to the user.

The field experiment was conducted over four months in 2018. The experimental design (Figure 1) captured whether there is a significant positive reaction to multiple, as-needed advisory messages; whether the structure of the message (level of simplicity of the message) matters; and which users are most likely to positively respond to such a message. We hypothesize that

Figure 1. Overview of the Experimental Design and Participant Allocation to the AI-Based Overdraft Prediction and Reminder Conditions



Notes. Experimental design and overview of participant allocation to the different conditions in the experiment. AI, artificial intelligence; ODP, overdraft prediction.

reminder alerts may have an impact on individuals with limited attention and that the design of the reminder messages, especially simplicity, may intensify this effect. We also conjecture that framing the email notifications differently, with a negative or positive valence, will have different effects on individuals who are prone to rational limited attention or ostrich behavior. Our hypothesis is that positive framing may trigger more reactions from those who avoid bad news: the ostriches.

After screening users into the sample using the algorithm, the 39,607 unique Mint ODP users (see Table 1 for descriptive statistics) were randomly assigned to one of four conditions: (a) the Control group that received no message, (b) the Base group that received a message previously used by Mint (Figure 2, Base), (c) the Simplified Avoid group that received a negatively framed simplified message (Figure 2, Simplified Avoid), and (d) the Simplified Save group that received a positively framed simplified message (Figure 2, Simplified Save). We describe the motivation for each of these rubrics in the literature review.

Of this sample, 22,775 were returning ODP users, and 16,832 (~42.5%) were first time ODP users. Before the experiment, the return users had received Mint ODP’s Base message at least once, whereas first timers received their first message during the experiment. We control for the first time effect in the main empirical analysis (including a propensity score matching analysis). As the

Table 1. Descriptive Statistics: Number of NSF Fees

Condition	Observations	Number of NSF fees assessed											
		One week after first reminder			Two weeks after first reminder			One month after first reminder			End of experiment (four months)		
		M	SD	Median	M	SD	Median	M	SD	Median	M	SD	Median
Control	10,972	0.91	1.65	0.00	1.56	2.53	1.00	2.91	4.22	2.00	8.08	11.41	4.00
Base	9,422	0.86	1.58	0.00	1.48	2.45	1.00	2.79	4.27	1.00	7.82	11.12	4.00
Simplified Avoid	9,725	0.83	1.54	0.00	1.41	2.30	0.00	2.63	3.79	1.00	7.36	10.45	4.00
Simplified Save	9,488	0.83	1.51	0.00	1.43	2.36	0.00	2.70	4.04	1.00	7.68	11.67	4.00

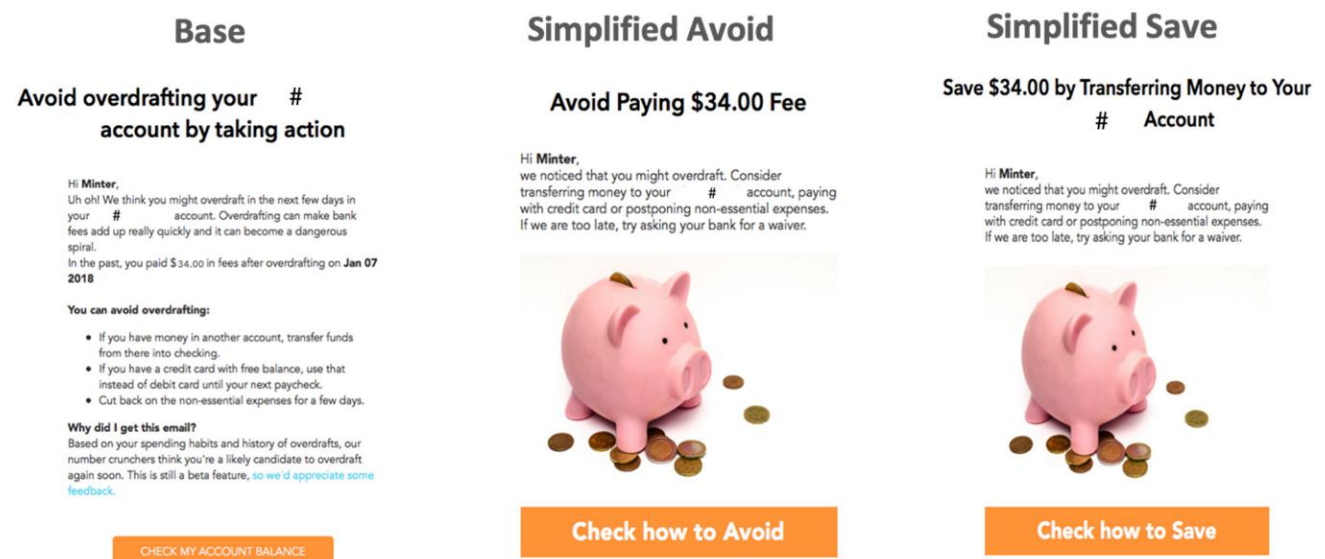
Note. The table shows the observations, means, standard deviations, and medians for the number of NSF fees assessed one week, two weeks, one month, after the first reminder was sent, and at four months.

sample is, by design, prone to overdrafts, the users are characterized, on average, by having significantly less available liquidity¹⁰ (during the experiment, only 38% of the sample had more than \$100 at each end of month), more debt, lower credit scores, and higher credit utilization than the overall Mint population (about 13% of users, on average, had a credit utilization rate below 50% and more than a total of \$1,000 in liquidity at the end of each month). Conditional on being identified by the algorithm and without treatment, a customer has a mean of 0.91 NSF one week after being identified, and 81% experienced at least one NSF/overdraft fee during the experiment.

We find that the Base message had a statistically significant positive effect in addressing inattention and reducing overdraft fees in the short term. Compared with the Control group, the mean number of NSF fees per person in the Base group was significantly reduced by 5.3% (from 0.91 to 0.86) one week after receiving the first reminder.¹¹ Simplifying the message substantially

enhanced its influence: it further reduced the cost of acquiring financial information and significantly reduced overdrafts by 9.25% (from 0.91 to 0.83) one week after receipt of the first notice compared with the control. Compared with the Base reminder, the Simplified Avoid version changed the impact of the message by 74.5% after one week and by 113.6% over the experiment period (four months). In dollar terms, for the four months of the experiment, the average saving for individuals receiving the Base reminder was \$9.33, and for the Simplified Avoid reminder, the average saving was \$25.16. The positive framing of the Simplified Save messages yielded an average saving of \$13.85 per user.¹²

Rather than test the impact of a one-time reminder, we investigate the efficacy of an algorithm that creates several reminders, as required, for the duration of the experiment. Each reminder treatment in our experiment may not just trigger a response to that intervention; it may also affect the frequency of subsequent reminders.

Figure 2. (Color online) Screenshots of Email Reminder Treatments: Base Message, Simplified Avoid (Negative Framing), and Simplified Save (Positive Framing)

Notes. The Base screen represents the original reminder that was sent to overdraft prediction algorithm users. The Simplified Avoid and Simplified Save versions were constructed in the experiment. Online Appendix D shows the interventions as a text.

Hence, we use time-to-event (TTE) analysis for recurring events to investigate the effect of each treatment on the risk of paying NSF/overdraft fees during the entire experiment.¹³ The results of the TTE semiparametric analysis also indicate that the simplified reminder significantly changed participants' behavior and lowered their risk of incurring overdraft fees for the duration of the experiment. Consistent with our previous results, the Simplified Avoid message outperformed the Simplified Save message in changing the risk for additional NSFs over time. We also conducted a counterfactual analysis using off-line contextual bandits (e.g., Agarwal et al. 2012) to see if we could find financial characteristics that might suggest using a mix of the three interventions would be better. We find no algorithmic strategy that would outperform the Simplified Avoid intervention based on the explored financial characteristics.

Next, we study the characteristics of those who were most responsive to the AI warnings and what mechanisms might have driven their responses. We find that the AI-driven solution mainly affects individuals/households with higher income (25th percentile and above) and fair to good credit scores (580–739). Among the actions that they could take after receiving a warning, one is to increase the funds in the account. We take advantage of the unique detailed transaction data to observe changes in balances before and during the experiment, looking at both liquid accounts (checking and savings accounts) and linked credit card accounts.¹⁴ We document a significant effect between receiving our notifications and the increase of credit card debt total balance among those who had not reached their credit limit,¹⁵ potentially as a mechanism to mitigate some of the costs.

Many in our sample had very high credit card utilization rates (the median credit utilization was 82%) and almost no liquidity, so if they chose to continue spending after the notification, they were likely to incur the overdraft fee. Our results indicate that an AI solution that increases attention is helpful but not always sufficient to avoid overdrafts. Less financially stable individuals may find it harder to act on the warning and avoid overdrawing. Our ability to differentiate (or decompose) in the experimental context between the three possible explanations for overdrawing an account—rationality with complete attention, rational limited attention, and ostrich behavior—is limited as financial constraints affect the range of possible responses in all three cases.

Our project contributes to the literature on AI-based financial advice, human–algorithm interaction, overdraft fees, financial hardship, and behavioral economics. To the best of our knowledge, we are the first to document the effect of using a machine learning algorithm to produce alerts designed to reduce overdraft fees. We contribute to the literature by quantifying the reaction of the individuals to the structure of the message, in particular, the simplicity effect and the framing effect.

Although we focus on learning how to reduce overdraft fees, understanding how individuals and households react to the different structures of messages delivered according to algorithm-based guidance in timely, as-needed reminders has applications beyond overdrawing accounts and even beyond finance.

Second, our findings contribute to the growing literature on mechanisms to improve the life of people who experience financial challenges by testing AI-based solutions. We also contribute to the human–computer interaction literature by documenting that different segments of the population may react differently to technological interventions and by documenting financial constraints that may affect an individual's ability to respond.

Third, we contribute to the methodological discussion about estimating the effect of an AI-based intervention that sends multiple, as-needed messages on the quality of financial decision making. We propose the use of the TTE method, which overcomes some of the endogeneity challenges. The ability to send multiple, as-needed messages is an important characteristic for potential AI solutions.¹⁶ We contribute to the literature on the intersection of nudging and AI-based systems literature (e.g., Bu et al. 2020, Allen et al. 2021, Shefrin 2021) by adding a system that can personalize the message based on financial context. Whereas, in our case, we don't find an advantage, we leave further exploration of this topic to future research with different designs.

In the literature review that follows, we elaborate on the overdraft phenomenon, the theoretical explanations, and the increasing use of digital and AI-based digital interventions. In Section 3, we describe the context of our study. These rubrics are followed by the research hypotheses in Section 4. The data and the experimental design are presented in Section 5. The results are in Section 6, and the summary and concluding remarks are in Section 7.

2. Literature Review

2.1. Overdrafts in General and in the United States

An overdraft occurs when a bank balance falls below zero. Banks have different protocols for overdrafts that range from declining the transaction that would cause a negative balance to charging fees and interest for the overdrawn amount. One such protocol is to set a maximum amount of credit a customer can borrow before overdraft fees and/or interest are charged. These fees and interest rates are, for the most part, fixed and relatively high. Some banks also charge periodic fees for maintaining an overdraft.

In the United States and Canada, many banks offer their customers overdraft protection, whereby the bank covers certain transactions that trigger an overdraft rather than declining the transaction, and it usually

charges an overdraft fee. When the bank does not accept a transaction, an NSF fee is charged irrespective of whether the customer is registered for the overdraft protection service. In addition to the NSF fee, banks usually charge an additional fee or interest for each day the account remains overdrawn. The overdraft protection service automatically approves costs, such as periodic transactions, automated teller machine (ATM) withdrawals, checks, debit transactions, and more. As of 2010, ATM cash withdrawals or debit transactions require the customer's explicit authorization.¹⁷ Small U.S. banks offer a similar service called bounce protection, which is usually managed by a third party according to a specific bank policy such that not all overdraft transactions are automatically covered.

Some banks also offer their customers overdraft transfer protection. This service connects a person's accounts or credit cards such that one account or card can be used to cover an overdraft on another. Customers pay a fee of \$5–\$10 per activation for this service.

Overdraft and NSF fees are pervasive in the retail banking system. In the United States, retail banks charge \$17 billion annually as NSFs, which account for approximately 6% of their total revenues (Stango and Zinman 2014) and for two thirds of their income accrued from fees. Elsewhere in the world, in February 2018, the UK Competition and Markets Authority imposed a regulation whereby personal current account holders are automatically enrolled in a program of reminders to notify them when they have exceeded or attempted to exceed a pre-agreed credit limit and will incur a charge (Grubb et al. 2025).

2.2. Rationality, Limited Attention, and Selective Attention

The literature offers several explanations for overdraft behavior. According to the rational models, people make consistent decisions after evaluating all available information and fully maximizing their total lifetime utility. With respect to overdrafts, individuals are assumed to make a conscious choice to pay the fees after thoroughly evaluating all the benefits of consumption, the cost of making an overdraft, and alternative sources of finance. Under the full-attention rational model, sending a reminder about overdraft fees would not affect an individual's decisions because they are assumed to be fully informed.

Limited attention or rational inattention models are based on the literature establishing that the process of acquiring information is complex (Franco et al. 2018, Bossaerts et al. 2019) or costly (direct or indirect costs), so people may make financial decisions after considering only a portion of the information available (Gabaix et al. 2006, Chetty et al. 2009, DellaVigna 2009, Sims 2010, Karlan et al. 2016, among others).¹⁸ These models are supported by extensive empirical literature (e.g., Hirshleifer and Teoh 2003, Ashraf et al. 2006, Stango

and Zinman 2014, Medina 2017).¹⁹ Rational limited attention may explain overdrafts insofar as some may find it costly to monitor their bank balances constantly or find it complicated to predict accurately the probability of an overdraft. In such cases, financial decisions may be based on partial information, resulting in overdraft fees. In instances of rational inattention, a reminder triggered by an algorithm that predicts overdrafts may reduce the costs of monitoring one's accounts. Furthermore, a less complex, clear message (as opposed to a longer, more technical text) may facilitate comprehension and be better at reducing fees. It is important to note that limited attention may be exacerbated in the digital era: obtaining information is relatively cheap yet, at the same time, may be overwhelming as individuals are constantly exposed to multiple and frequent stimuli (e.g., Cowan et al. 2005, Lindgaard et al. 2006, Benartzi and Lehrer 2015).

A third possible explanation relies on information-dependent utility models, such as those by Köszegi and Rabin (2009) and Andries and Haddad (2019). These researchers posit that individuals choose to ignore relevant information even though this can result in a cost. A behavioral explanation relevant to our inquiry is the ostrich bias effect defined as the avoidance of acquiring information regarding risky or unpleasant situations (Galai and Sade 2006, Ofir and Wiener 2012). Phrased metaphorically, individuals could be willing to pay a price for the bliss of ignorance. The ostrich effect may propel individuals to eschew gathering information about the balance of their account that could help them avoid overdrafts and paying penalties. We expect individuals who act as ostriches not to respond to the reminders as they are likely to entertain the vain hope that if they ignore the information, the menace will disappear. Among this population, responsiveness to reminders might be correlated with income as the latter may be negatively correlated with financial concerns (e.g., Olafsson and Pagel 2017). Furthermore, to the extent that ostriches accept economic pain to avoid negative news, they may respond better to positive messages, phrased in terms of saving, than to the negative framing in the form of warnings about penalty fees.

2.3. Digital and AI Interventions

As digital technology evolves, there has been a significant increase in the use of digital interventions aimed at influencing decision making. AI-based interventions, in particular, are gaining popularity across various sectors, including healthcare, organization management, transportation, law, marketing, and finance. Across academic disciplines, there are many research questions in the field of digital and AI interventions. These include methodological system design, types of algorithms employed, ethical considerations surrounding their use, and the focus of this project: algo-human interaction.²⁰

We contribute to this literature by exploring the effects of an assistive AI system that provides guidance and employs recurring interventions in consumer finance. We not only analyze the outcomes of a large-scale field experiment but also provide evidence of the underlying mechanism. We contribute to the discussion about the empirical methodology of such AI-based experiments by proposing that time to treatment initiation survival analysis be used.

3. Setting

We use Mint, a large, web-based PFM application that monitors bank transactions, credit cards, investments, and loans. Mint's customers connect their credit cards and bank accounts to the application, which generates an integrated personal interface. One of Mint's advanced services is ODP, an algorithm that computes an individual's likelihood of initiating a transaction that exceeds the individual's available funds in the coming week. This service was initially (and during our experiment period) offered to the customers of the top three U.S. banks. If the algorithm calculated a score above a given threshold,²¹ an email was sent to the user (see Figure 2, Base). The AI algorithm predicts these events a week ahead, giving customers time to take action to avoid an overdraft. Recurring messages were sent as needed.²² A unique aspect of our experimental setting is that the action in response to the algorithm suggestion has to occur in a timely manner. This is an important characteristic that increases our confidence that the alerts triggered the actions. The algorithm is a gradient-boosting decision tree model with periodic retraining.²³ The calculation of the score involves a set of features that can be conceptually divided into three groups related to (a) user profile (e.g., the number of accounts owned, periodic income), (b) balance features (e.g., cash-flow behavior and balance trends), and (c) transactions (e.g., previous NSF fees, number of small debit and credit transactions, frequency of purchases). Unsurprisingly, the sample population tested is characterized by substantially less available funds, less savings, fewer assets, more debt, and higher credit utilization, on average, compared with the overall Mint population (see Online Figure 4 and Online Table 8 for credit scores and credit card utilization rate distributions). Hence, our sample is not representative of the North American population or of Mint users.²⁴ However, the focus on this population is very relevant as we want to study the impact of information architecture and of sending reminders as needed in complex financial contexts on individuals who are more likely to pay overdraft fees.²⁵

4. Motivation for the Research Questions

Our project builds on the behavioral economics studies that show messages can shape people's decisions and mitigate limited attention in different contexts. Current

technology can be used to send as-needed messages at a time calibrated to the need for action, which creates different dynamics and prompts new research questions. As our aim is to better understand AI-human interaction, an important focus of our randomized field experiment is to learn whether the AI-based structure and framing of recurring email reminder messages make a difference in terms of overdraft fees. We focus on recurring events of (a) a simple versus more elaborate message and (b) a positive versus negative framing. We elaborate on the motivation and the hypotheses for each of these manipulations.

4.1. Simplicity

Whereas receiving a reminder, in and of itself, may mitigate the limited-attention problem by reducing information cost and complexity, the phrasing of the message can further influence these outcomes. Given the abundance of screen-based stimuli, simplicity may be an important aspect of the information architecture of a notification (e.g., Tuch et al. 2010, Benartzi and Lehrer 2015). Cowan et al. (2005) document that individuals tend to form impressions and make decisions based on information presented and processed in a very short time and with very little attention investment. In our experiment, illustrated in Figure 2, we compare the effect of a message designed and used by Mint ODP (Base) compared with that of a simpler message created for the purposes of this research. The simplification of the message involves (a) shortening the text, (b) giving the fee amount more salience by displaying it as part of the headline, (c) presenting related art that is associated with savings, (d) increasing the size of the "explore account" link, and (e) creating a related actionable text. For robustness, to confirm that the simplified version is indeed simplified, we ran an online experiment survey using Amazon's Mturk platform²⁶ with an adult sample of 107 participants from the United States. The results confirm that the simplification of the reminder was perceived as intended by a significant margin; see Online Appendix Manipulation Tests for details.

We hypothesize that individuals with a tendency toward rational limited attention will respond positively to reminders. Further, we hypothesize that, compared with long text messages, short and simple messages will be more effective as they further reduce the cost of gathering and processing information. We expect that the level of simplicity of the reminders will have no effect on individuals prone to ostrich bias, who may not even open them. We also hypothesize that an increase in attention triggered by a reminder will result in actions that will reduce the mean number of NSF fees.

4.2. Positive vs. Negative Framing

The concept of framing is introduced and developed by Tversky and Kahneman (1981), who show that a

negative framing of a message, emphasizing a potential loss, triggers a different response than a positive framing with a focus on potential benefits.²⁷ A rich discourse has evolved on the positive versus negative valence of messages: some studies show that a positive utterance has a greater impact (e.g., Levin and Gaeth 1988, among others), whereas others maintain that responses to a negative utterance are more acute (e.g., Block and Keller 1995, among others). A framing typology for the different situations explored is developed by Levin et al. (1998).²⁸ Our project focuses on what the persuasion literature terms “goal framing,” which differentiates between emphasizing the benefit of obtaining a positive outcome versus avoiding a negative consequence. To test the goal-framing effect, we conduct the following manipulation: keeping the body of the simplified message unchanged, we use two versions of the heading: one framed negatively (Avoid) and the other positively (Save). Again, for robustness, we test our design using Amazon’s Mturk platform. The negative framing was significantly perceived as more negative than the positive framing. Although a substantial part of the literature on goal framing suggests that negative framing has a stronger impact, within the setting of our experiment, one might hypothesize that ostriches are more affected by the positive message since they tend to avoid negative news.

4.3. Recurring Messages with Framing

There is a large academic literature on positive versus negative framing, yet because recent technology makes it possible to send multiple electronic reminders as-needed at the time of need, this may create different dynamics. Even though there is growing evidence of the efficacy of reminders in general (Karlan et al. 2016, Medina 2017, Alan et al. 2018, Bursztyn et al. 2018, DellaVigna and Linos 2020, Grubb et al. 2025),²⁹ to the best of our knowledge, there is very little research on the use of recurring reminders with different framings and structures in consumer finance. Given the rise of financial technology and the increased use of AI, we need to discuss different methods of estimation to understand the outcomes of the intersection of recurring, as-needed messages with framing.

4.4. Income and Reminders

Individuals with different income levels may respond differently to reminder treatments and differently again to digital reminders (Mani et al. 2013, Haran Rosen and Sade 2022). One conjecture is that low-income individuals react more as paying a fee takes up a larger portion of their income. However, low-income individuals may not have the resources they need to take action: they may not have better credit alternatives or may have a higher cognitive load due to other hardships, rendering them less able to act on the warning. Given this, income is an important factor on which to focus.

We aim to (a) establish whether the recurring, as-needed reminders have a different effect on fees for people at different income levels; (b) provide additional evidence to interpret the different channels that may affect our results, including financial constraints; and (c) relate our findings to the limited-attention and ostrich-effect literature.

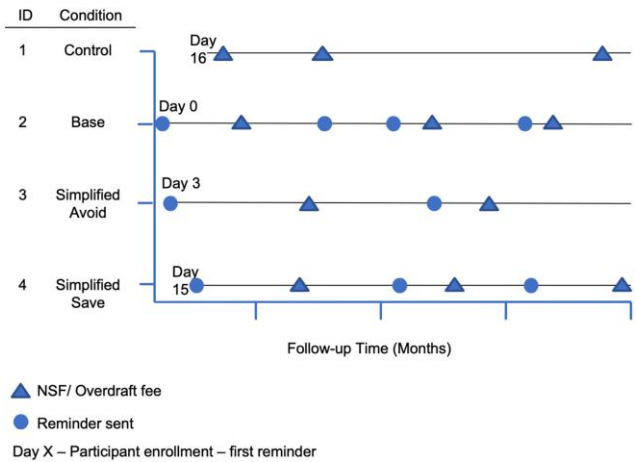
4.5. Credit Score and Reminders

Individuals with different credit scores may respond differently to treatments and to digital reminders (e.g., Bracha and Meier 2015, L’Esperance 2023). As credit scores are correlated with personal characteristics, one conjecture is that low credit score individuals should react more, and those with higher credit scores might need fewer reminders. Another conjecture is that those with high credit scores react more as they may find it easier to act upon the warning via financial channels and prevent the hazard. Given this, in addition to income, credit score is an important factor upon which to focus. Because, to the best of our knowledge, there is no significant documentation on the correlation between credit score and the responsiveness to recurring reminders and to different ways of delivering them, we aim to provide a better understanding of individuals’ behavior in such circumstances and relate our findings to the limited-attention and ostrich-effect literature.

5. Data and Experimental Design

The experiment was carried out over four months in 2018³⁰ and had two stages as illustrated in Figure 1. First, the Mint ODP algorithm calculated a score for the Mint user sample. If the score exceeded a threshold, the user was randomly assigned to one of the four experimental groups:³¹ (a) the Control group: users do not receive any reminder message; (b) the Base group: users receive the original message (Figure 2, Base); (c) the Simplified Avoid group: users receive a message with a simplified negative text framing (Figure 2, Simplified Avoid), and (d) the Simplified Save group: users receive a reminder with a simplified positive text framing (Figure 2, Simplified Save). Users exposed to one of these treatments did not receive any additional messages during the following week. Each user could receive up to four messages over one month. All messages sent to the same user were of the same type. Overall, 39,607 unique users participated in the experiment. During the first month of the experiment, new participants could be assigned to one of the treatments, but no additional participants were included after the first month. We observed all the participants in the sample for at least three additional months and for up to four months total.³² An illustration of the time frame is presented in Figure 3.

Figure 3. (Color online) Illustration Timeline of Participant Follow-Up Periods Across Experimental Conditions



Notes. Illustration of the follow-up period in months for the participants in different conditions. Over the first month, new participants could be assigned, and we subsequently followed them to the end of the experiment and for at least three more months.

Of the 39,607 participants, 10,972 were randomly assigned to the Control group, 9,422 to the Base group (the original message), 9,725 to the Simplified Avoid group, and 9,488 to the Simplified Save group. Over the four months of the experiment, 81% of the participants incurred at least one NSF or overdraft fee; the average number of fees for the total sample stood at 7.8 ($SD = 11.01$) with a median of 4.

The average annual income for the sample was approximately \$108,000 ($SD = \$1.02M$) with a median of \$53,200. We divided income into six groups according to the U.S. 2018 individual income using different quantiles. We were able to obtain credit report data for about 53% of the sample. We divided credit scores into five groups according to Ascent report divisions. On average, the credit score was 608, and the median was 610. As shown in Online Table 9, about 77% of the population in our sample, conditional on observing their credit scores, had a credit score between 300 and 669, 22% between 670 and 799, and only 1% in our sample had a credit score between 800 and 850.

5.1. The Experiment and the Data: Advantages and Challenges

Using unique proprietary data and a field experiment offers several advantages. (a) Our access to Mint's data enabled us to observe and investigate detailed data on many users: specifically, we could observe the sample's bank activity, credit cards, investments, loans, and transactions, including overdraft and NSF fees. (b) Within the framework of a controlled field experiment, we knew exactly when each reminder was sent and could monitor the participants' subsequent behavior

and potential barriers to action directly. (c) Participants were randomly assigned to experimental and control conditions, which enabled us to identify the causal effect of receiving a message and the marginal effect of the different versions. (d) The Mint application enabled us to monitor user engagement following each reminder by observing and measuring the open rates and the click-to-explore rates. (e) We could investigate the effect of recurring reminder events as some of the participants received more than one reminder. (f) We could test the combined effect of the financial innovation introduced by the application (the algorithm that predicts overdrafts) with the various phrasings of reminders in an endeavor to find effective means to reduce overdraft fees. (g) Beyond observing the outcome, the level of fees, we can investigate different channels of actions that occur after the warning is delivered.

Notwithstanding the merits of our design, our data and experiment present several challenges. (a) Individuals who opt to use a financial application may not be representative of the general population³³ as they may have higher levels of awareness and attention. Arguably, our sample is selected based on user interest and actions. Yet, whereas this could be relevant for comparing the app users to the general population, we employ random assignment within the experiment. (b) The experiment was conducted over four months, and the targeted effects were monitored during that period alone. Any subsequent activity is, thus, beyond the scope of this study. It has been suggested (e.g., Stango and Zinman 2014) that the effect of notifications decreases over time. One advantage that our setting affords is that the experiment also explores the effect of reminders on first time compared with return users. This difference, in our opinion, partially offsets the limitation associated with the short duration of the experiment. (c) Prior to the experiment, in the event of an overdraft risk, Mint ODP sent a long version of the notification via email. This means that some of our participants had previously received the Base message, whereas the rest received an experimental version for the first time. The effect of reminders on first time users may differ on account of novelty, and in addition, that population may inherently be at a lower risk of an overdraft; that is, this could explain why reminders were not previously sent. To address the potential difference between these two populations, we control for the first time effect, and we add propensity score matching analysis in the robustness section. (d) Our income analysis³⁴ is based on accounts of users registered on the application. It is possible that a user may not have reported all of the user's accounts, so we cannot guarantee that the information gleaned from the application captures each user's total income. Although including only partial information contravenes most people's motivation for using the application, we must be cautious in that regard.³⁵ To further address this concern, we also

analyzed results at different income levels for customers who linked at least two bank accounts as an attestation for a more complete picture. We also acknowledge and address that, in our analysis, there is no clear distinction between individual and partnered household incomes, and we analyze both. (e) One of our analyses that aims to understand better who is being affected focuses on overdraft reductions and credit scores. The credit score information is based on information obtained from linked credit card accounts and credit report data. Whereas our unique setting enables us to link this information to a subsample of our sample population, we can't do this for the whole sample. Still, this subsample holds significant value in enhancing our understanding of the mechanisms and how individuals respond in such situations.

6. Results

6.1. Sending Different AI-Based Reminders and Overdraft Fee Payments

The impact of reminders on overdraft fees is measured in several ways. We lay these out in Table 2. We initially report in the text the results of the following two measures: (a) time periods after the first reminder (TAFs)—we report the effect one week after participants received the first experimental version with the aim of measuring the immediate effect of the algorithm prediction³⁶—and (b) total time of the experiment (TT) for the four-month duration of the experiment, starting from the beginning.³⁷

We start by examining the effect of sending a reminder on the number of overdraft charges paid by the participants (Tables 2 and 3).³⁸ Compared with the Control group, sending any kind of reminder showed a significant reduction in NSF fees. The Base ODP reminder showed an NSF fee reduction of 5.3% (from 0.91 to 0.86) in the mean number of NSF fees per individual one week after the first reminder and 3.3% (from 8.08 to 7.82) at four months.

Next, we document the effect of simplifying the message. The Simplified Avoid message, which kept the negative valence of the Base message, increased attention and actions: it not only significantly affected the open rate of the emails and the click-to-explore (opening the link that enables further information and actions),

but it reduced the mean number of NSF fees per person by a significant 9.25% one week after the first reminder and by 8.9% for the full four months.³⁹

When changing the simplified message from Avoid (negative) to Save (positive), we find that the NSF fee-related effect size was 9.44% after one week and 4.9% for the full experiment (four months). The data suggest that the Simplified Save version is more susceptible to value decrease over the extended period, that is, at least one month.⁴⁰

6.2. Addressing Endogeneity Resulting from the Multiple, As-Needed Messages

One important characteristic of the AI solution is that it sends alerts as needed. We explore whether multiple, as-needed interventions affect decision making over time. Because the number of interventions sent and the timing of the interventions is based on the overdrawing algorithmic likelihood score, there is an inherent endogeneity challenge in estimating whether one or more interventions are needed. Also, each of the treatments implemented may not only trigger an action related to a specific event but also affect the frequency of further treatments. This is the credit assignment problem documented in the literature (e.g., Sleeman et al. 1982).

To address these challenges in the empirical analysis, in addition to reporting the cross-sectional analysis, we follow Prentice et al. (1981), who develop a variation on the traditional survival analysis, and apply conditional risk set models of TTE analysis.⁴¹ This procedure enables us to analyze the potential differences in NSF fees between the groups, accounting for the time at which the event (NSF fee) occurred. These models incorporate time-dependent covariates (for each participant) with recurring and ordered events (NSF fees over time in which the risk for the next NSF fee event is only for those who experienced an event on the previous stratum). We use the semiparametric identification of the Cox model to relax the predefined distribution of the hazard to pay an NSF fee. In such an analysis, the dependent variables are both the number of NSF fees and the time at which the events occurred. Hence, our hazard estimation function

Table 2. Summary Statistics: Treatment Effects

Condition	NSF fees (one week after reminder) (1)	NSF fees (four months) (2)	Open rate (3)	Click-to-explore rate (4)
Base	−5.3%***	−3.3%*		
Simplified Avoid	−9.25%***	−8.9%***	12.1%***	78.4%***
Simplified Save	−9.44%***	−4.9%***	11%***	7.8%
Control	0.91	8.08		
Base			27.5%	1.4%

Notes. Columns (1) and (2) show average treatment change compared with the Control group in number of NSF fees one week after receiving the first reminder and for the four-month experiment. Columns (3) and (4) show the average open rates and click-to-explore rates for the simplified alternatives (Avoid, Save) compared with the Base (original) message. Open rates and click-to-explore rate are calculated on any reminder in the experiment. The Control and Base values below the line are means.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 3. Tobit Regression for Log(1+ NSF Period) for the Mean Number of NSF Fees One Week, Two Weeks, and One Month After First Reminder and for the Four Months of the Experiment: Control vs. Base and Base vs. Simplified Avoid Group

Variable	Base vs. Control group				Simplified Avoid vs. Base group				Simplified Avoid vs. Simplified Save group			
	Full experiment (four months)	One month	Two weeks	One week	Full experiment (four months)	One month	Two weeks	One week	Full Experiment (four months)	One month	Two weeks	One week
Base	−0.0316*	−0.0386**	−0.0327**	−0.0283**								
	−0.0192	−0.0159	−0.0137	−0.0116								
First timers	−0.807***	−0.533***	−0.393***	−0.271***	−0.761***	−0.500***	−0.367***	−0.246***	−0.789***	−0.516***	−0.361***	−0.233***
	−0.0196	−0.0154	−0.0127	−0.0106	−0.0209	−0.0165	−0.0136	−0.0113	−0.0209	−0.0164	−0.0136	−0.0112
Base × First timers	0.047	0.033	0.0265	0.0248								
	−0.0286	−0.0226	−0.0187	−0.0155								
Yearly income	0.0952***	0.0594***	0.0536***	0.0417***	0.0794***	0.0534***	0.0499***	0.0384***	0.0273	0.0281	0.0219	−0.00092
	−0.00787	−0.00632	−0.00536	−0.0045	−0.00827	−0.00666	−0.0057	−0.00476	−0.0293	−0.0231	−0.0191	−0.0158
Simplified Avoid					−0.0530***	−0.0399**	−0.0345**	−0.0199*	−0.0502**	−0.0298*	−0.016	−0.00298
					−0.02	−0.0165	−0.0141	−0.0119	−0.0201	−0.0166	−0.0142	−0.0119
Simplified Avoid × First timers					−0.000569	0.0129	0.0282	0.0126	0.0754***	0.0419***	0.0359***	0.0271***
					−0.0294	−0.0232	−0.0191	−0.0159	−0.00836	−0.00678	−0.00568	−0.0047
N	20,394	20,394	20,394	20,394	19,147	19,147	19,147	19,147	19,213	19,213	19,213	19,213

Notes. The table shows Tobit regressions and marginal effects; delta-method standard errors are in parentheses. Dependent variables are NSF fees assessed one week, two weeks, and one month after the first reminder sent, and the four months of the experiment. *Base* is the condition in which the underlying variable is the Control condition. *First timers* is a dummy variable for new overdraft prediction users. *Base × First timers* is an interaction variable. *Yearly income* is the log(yearly income). *Simplified Avoid* is the condition in which the underlying variable is the Base condition. *Simplified Avoid × First timers* is an interaction variable.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 4. TTE Analysis and Semiparametric Cox Model: Base vs. Control Group and Simplified Avoid vs. Control Group

Variable	Base vs. Control group (1) TTE-PWP-Cox	Simplified Avoid vs. Control group (2) TTE-PWP-Cox	Simplified Avoid vs. Simplified Save group (3) TTE-PWP-Cox
<i>Base</i>	–0.00875 –0.00825		
<i>Yearly income</i>	1.66E-09 –3.98E-09	–2.58E-10 –3.48E-09	6.50E-09 –7.35E-09
<i>First timers</i>	–0.208*** –0.0131	–0.206*** –0.0131	–0.210*** –0.0144
<i>Base × First timers</i>	0.00559 –0.0185		
<i>Simplified Avoid</i>		–0.0183** –0.00824	–0.0192** –0.00864
<i>Simplified Avoid × First timers</i>		–0.00858 –0.0187	0.00169 –0.0195
<i>N</i>	114,604	113,546	102,724

Notes. The table shows the results of the TTE analysis and PWP (Prentice-Williams-Peterson)-Cox semiparametric regression; robust standard errors are in parentheses. Dependent variables are both NSF fees and time to NSF event (the hazard for NSF). *First timers* is a dummy variable for new Mint overdraft prediction users. *Simplified Avoid* is the condition in which the underlying variable is the Simplified Save condition. *Yearly income* is the log (yearly income). *Simplified Avoid × First timers* is the interaction variable.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

is as follows:

$$\lambda_{ik}(t) = \lambda_{0k}(t) * e^{X_{ik}\beta} \quad (1)$$

Here, $\lambda_{ik}(t)$ is the hazard function for individual i for event k (NSF fee occurrence) at time t . As mentioned above, we do not assume a base distribution of the hazard function λ . β is a vector that includes the following: the treatment dummy variables, *Base* for when the underlying alternative is no message (Control condition) and *Simplified Avoid* when the underlying alternative is the original Base message; *First* is a dummy for first timers, and *Base_First/Simplified_First* are the interaction variables between the two. In both cases, *Yearly income* represents the participant's yearly income. The standard errors are robust.

In our specification, the time periods are days. We measure time to event k from time t , where t stands for the time of the previous event or of the participant's entry to the experiment. Insofar as the unit of time is set as one day, the TTE analysis treats all NSF or overdraft fees paid during a single day as one event. The results are displayed in Table 4. The Base version significantly reduced the number of NSF fees paid during all the time periods investigated (one week, two weeks, one month, and four months) for both TAF and TT estimations. Simplifying the message significantly increased its impact compared with the Base version for all time periods in both TAF and TT estimations. These results likewise emerged as significant when controlling for first time effects and income.⁴² Within the TTE analysis, we find that the Base alternative reduced the number of NSF fees, albeit not significantly; nevertheless, the reduction achieved by the simplified version is significant.

6.3. Who Is Affected by the Interventions and the Channel They Use

To reduce the NSF fees, individuals must pay attention and be able to act upon the new information. To better understand those affected by the information, we take advantage of the unique Mint data, and we explore distinguishable financial characteristics. We first compare the impact of receiving reminders for each of six income groups relative to the Control group and then pursue a similar process for credit score groups.

For our income analysis, we divide the sample into the following percentile groups:⁴³ (a) total annual income of less than \$10,000; (b) up to the 25th percentile, \$10,000–\$20,000; (c) up to the 50th percentile, \$20,000–\$40,000; (d) up to the 75th percentile, \$40,000–\$70,000; (e) up to the 90th percentile, \$70,000–\$115,000, and (f) an annual income of \$115,000 and above.

Within the experimental sample—which, as already noted, comprises individuals with a higher probability of overdrawing an account than the general population—we observe that participants with a relatively higher total annual income (as estimated on a full year prior to the experimental period) pay more NSF fees. In the lowest income level (less than \$10,000), the mean total number of NSF fees paid stands at 4.5 and 4.77 for the control and treatment populations, respectively. As income rises, this number steadily and significantly increases, culminating in the respective means of 10.46 and 9.43 for the highest income group. Table 5 displays the results.⁴⁴

The income analysis reveals that sending reminders to individuals with relatively higher income—those in the 25th income percentile and above—significantly reduced their number of overdraft charges, at least in the short term,⁴⁵ and the effect is the same for the total

Table 5. Tobit Regressions for Number of NSF Fees One Week After the First Reminder and Four Months of the Experiment: Reminder vs. No Reminder by Income Group

Variable	Four months of experiment					One week after reminder				
	Up to \$10,000 (up to 25th percentile)	\$10,000–\$20,000 (up to 50th percentile)	\$20,000–\$40,000 (up to 75th percentile)	\$40,000–\$70,000 (up to 90th percentile)	\$70,000–\$115,000 and above (above 90th percentile)	Up to \$10,000 (up to 25th percentile)	\$10,000–\$20,000 (up to 50th percentile)	\$20,000–\$40,000 (up to 75th percentile)	\$40,000–\$70,000 (up to 90th percentile)	\$70,000–\$115,000 and above (above 90th percentile)
<i>Reminder</i>	–0.0123	0.0263	–0.0478*	–0.0332	–0.0768**	–0.056	–0.00588	–0.0455***	–0.0377**	–0.0518**
<i>First timers</i>	–0.109	–0.0543	–0.0288	–0.0279	–0.0361	–0.0606	–0.0319	–0.0168	–0.017	–0.0254
	–0.745***	–0.508***	–0.674***	–0.771***	–0.900***	–0.185***	–0.164***	–0.243***	–0.250***	–0.312***
	–0.115	–0.0634	–0.0361	–0.0367	–0.047	–0.0629	–0.0348	–0.0193	–0.0203	–0.0247
<i>Reminder × First timers</i>	0.0399	–0.0138	0.0710*	–0.0146	0.0562	0.0605	–0.00521	0.0578**	0.00426	0.0598**
	–0.138	–0.076	–0.0427	–0.0428	–0.055	–0.074	–0.041	–0.0228	–0.0236	–0.0336
<i>N</i>	827	2,818	10,178	11,508	7,435	827	2,818	10,178	11,508	6,552

Notes. The table shows the tobit regression analysis of the number of NSF fees one week after the first reminder and for the four months of the experiment for the six income groups. Marginal effects are presented, and delta-method standard errors are in parentheses. *Reminder* refers to any kind of reminder, and the underlying variable is the Control condition (no reminders). *First timers* is a dummy variable for new overdraft prediction users. *Reminder × First timers* is the interaction variable.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

experiment duration among individuals in the 75th percentile and above. Overall, the influence of the reminders increases with yearly income. These results differ from the Stango and Zinman (2014) findings, which show a greater influence of salience shocks on overdraft fees for lower income individuals. Nevertheless, it is consistent with the literature documenting that those with lower income are less prone to act to their own benefit once a related mechanism that reduces the costs of obtaining needed financial information is provided to them (e.g., Haran Rosen and Sade 2022).⁴⁶ This connection raises the question of why, in our contexts, are not all income groups influenced by the reminders in the same magnitude and manner.

We now turn to our credit score analysis. For this, we divide the sample into the common five credit score groups: (a) 300–579, (b) 580–669, (c) 670–739, (d) 740–799, (e) 800–850. Online Table 9 shows the distribution of NSF fees to the different credit score groups and describes our sample relative to the general American population distribution.⁴⁷

Online Figure 4 shows the expected strong negative correlation between credit score and the number of NSF fees. We investigated the differences in the number of NSF fees paid between the control group and the treatment population (i.e., those who received any version of intervention) with respect to credit score groups (Online Table 10).

It seems that there is a positive correlation between the interventions and the reduction of NSF fees across credit score groups except for the highest group.⁴⁸ Because we obtain only about 53% of the experiment population users' credit scores and split the analysis into five subgroups, the statistical power for each group is reduced. Sending reminders to individuals statistically significantly reduced NSF payments for the 580–669 credit score group for the four-month duration of the experiment and had a marginally significant effect over the 670–739 credit score group. We find that the interventions were effective for the group of individuals in the lower credit score group (300–579) one week after the first intervention, but this effect does not hold for the full experiment duration. This may relate to the longer term ability of this group to react as they might not have access to other financing tools to manage their income shortfalls. To the best of our knowledge, we uniquely report overdraft fee payments across credit score groups and the effect of reminders in general and AI-generated reminders in particular on these groups.

In addition to the financial characteristics, we also seek insight into the channel or mechanism that individuals use to avoid or reduce paying overdraft fees after receiving the algorithm-generated messages. We focus on three possible channels. Two of them are related to financial decisions and the third to purchasing decisions, and these channels are not mutually exclusive.⁴⁹

The first channel is to seek different methods of financing. In our setting, we study changes in credit card debt. The second channel is to increase liquidity in one account by transferring funds from a different account, and the third channel is to postpone or avoid discretionary expenses. To determine whether users are using another financing mechanism, such as credit cards, to avoid paying fees, we calculate the end-of-month total credit card balance for all of the credit card users linked to the app. We then compute the following estimates: (2) the first differences of the credit card total balance difference for each month and (3) the total balance difference for the full experiment duration. We estimate the regression specification:

$$\Delta_{t-t-1} \text{Credit Card Balance}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i, \quad (2)$$

$$\Delta_{t_{\text{last}}-t_{\text{first}}} \text{Credit Card Balance}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i, \quad (3)$$

where *Reminder* refers to anyone who got the intervention and *Income* represents a log for the yearly income of individual *i*. *t* represents a month time frame. *t_{last}* represents the end of the month of the end of the experiment, and *t_{first}* is the first month of the experiment. ε_i represents the error term for individual *i*.⁵⁰

As a significant portion of the experimental population received more than one reminder, we explore whether users who received reminders increased their credit card balance after receiving the messages compared with the control group, who did not receive any messages.

We conducted this exercise for the full sample.⁵¹ Users who received the simplified version marginally significantly increased their credit card balance during the first month of the experiment. Delving deeper into who is influenced, we find that, when conditioning on users who didn't completely max their credit limit (removing just the 100th percentile or those who fully used their credit limits), we observe a significant increase in credit card balances. This result is also found when taking into consideration users in the 580–740 credit score cohorts. Even though we find this significant difference in the first experiment month, we do not observe the same effect on credit card balances when comparing the difference over the full period of the experiment.⁵²

The second channel is transferring money into the account. Money may come from different accounts owned by the same individual/household (linked or not linked to the app) or from an external source (e.g., family support). Because we can't be sure if a transfer was made by the same household from a different account or by others, we analyze a deposit from any source. To do that, we calculate the overall monthly inflows for the user-linked accounts.

We then compute (4) the first differences of the inflow difference for each month to the previous month and (5) the inflows for the full experiment duration. The specification is

$$\Delta_{t-t-1} \text{Money inflow}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i, \quad (4)$$

$$\Delta_{t_{\text{last}}-t_{\text{first}}} \text{Money inflow}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i, \quad (5)$$

where *t* represents a month time frame. *t_{last}* represents the end month of the end of the experiment month, and *t_{first}* is the first month of the experiment. ε_i represents the error term for individual *i*.⁵³ We find no consistent significant correlation between the reminders and money inflow differences during the experiment.

Lastly, to address the third channel, we see if users postponed some expenditures. Because this would require observing something that did not happen, we explore a proxy by looking for a correlation between overall expense differences observed during the months of the experiment and throughout the full experiment. To do so, we calculate the overall monthly money outflows for the user-linked accounts. We then compute the first differences of the outflow difference for each month and the outflows at the beginning versus at the end of the experiment. The specification is

$$\Delta_{t-t-1} \text{Money outflow}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i, \quad (6)$$

$$\Delta_{t_{\text{last}}-t_{\text{first}}} \text{Money outflow}_i = \alpha + \beta_1 \text{Reminder}_i + \beta_2 \text{Income}_i + \varepsilon_i. \quad (7)$$

We find no significant difference in postponing expenses across months between our user group, who received reminders, and the control group.⁵⁴

6.4. Different Compelling Possible Explanations Provided in the Literature

According to our hypothesis and the academic literature, sending a reminder makes a difference, in and of itself, at least in the short term. It significantly helps address the inattention that, in a financial context, may result in overdraft fees. In the experiment, reminders increase individuals' attention to their financial situations and reduce the NSF charges. Simplifying the message substantially enhances this impact, changing participants' behavior by further reducing the cost of acquiring salient financial information.

Another attestation to limited attention can be found by decomposing the analysis into overdraft and NSF fees. Overdraft fees can be classified as a financing mechanism, whereas NSF fees are not since the transaction was not approved. As described above, the two fees are addressed interchangeably since, in many cases, identifying the declined transaction out of the imposed

NSF fee is not clear. Also, for some banks, the fee transaction description is mixed.

Even so, we show significant and similar effects on both types of fees.⁵⁵ Using textual analysis, we differentiate between the two kinds. In our sample and during the experiment, among those who paid fees, 74% paid overdrafts, 66% paid NSFs, and about 40% paid both types. By reducing NSF fees, we are just saving money for the user and not changing any financing decision that might be implied by paying an overdraft fee. This can be seen as another indication that limited attention is what is observed. To address potential ostrich behavior, we frame messages more positively to overcome the aversion to negative information. Given that ostrich behavior is a way to avoid information about hazards and negative consequences and given that financial hardship influences the ability to act to reduce the likelihood of an NSF fee, we explore whether those who are in a more challenging financial situation react differently to positive framing.

We employ and test two separate, different financial conditions of a potential challenging situation: (a) users who we observed as having a significant negative trend on their total account balances six months prior to the experiment and (b) users who are in lower income groups. In both cases, we introduce an intervention that includes a positively framed message. We change “avoid” to “save” (see Figure 2 for the simplified positive intervention). We measure if, conditional on being in either one of the two conditions, users react more to such positively framed warnings compared with the negatively framed ones. The reaction is measured by observing overdraft reductions and by differences in open rates (i.e., opening the email) that enable us to explore awareness *per se*.⁵⁶ For the estimation of the first financial condition identification, we use two-stage regression. The first stage is a time series trend regression to identify users who have negative balance trends on their total balance in the linked accounts six months prior to the beginning of the experiment. To those who had at least a marginally significant negative trend, we gave the value one and zero otherwise. Stage two explored whether those with a negative trend reacted differently to the reminder in both overdraft payments and open rates.⁵⁷

Overall, users with a negative observed trend opened the sent reminders significantly more than those without a significant negative trend. Also, they open the simplified negative framing more than the positive; even though we observe the willingness to open the reminders, they make no significant difference to whether they overdraw their accounts.

In addition to the trend analysis, we test whether lower income individuals display more ostrich behavior and whether the framing of the message mitigates such behavior. We find that only the second income group of

up to the 25th percentile experienced significant differences in open rates with higher open rates found for the Simplified Avoid version. The difference in open rates for all other income groups was not significant. We observed no consistent differences between the Avoid and Save versions in terms of the reduction in the number of overdraft fees paid. Hence, for both the trend analysis and the income-level analysis, we find no clear or consistent evidence that we can specifically identify or mitigate ostrich-driven behavior. Although we could not clearly identify such behavior, that does not necessarily imply that such an explanation or behavior is not present. The Mint population of individuals/households who decided to use a PFM may be less susceptible to ostrich behavior. The last suggested explanation is the fully rational explanation, by which people pay overdraft fees as a financing mechanism, and the cost of the fee is lower than the value they get by completing the transaction.

Our sample population includes a large portion of users who have limited financial recourse when they receive a reminder alert. By design, getting the algorithmic alert means that the individual/household is predicted to face a cash-flow challenge and may not have enough liquidity (cash amounts on liquid accounts) for future transactions. Additionally, other financing resources are also scarce. Based on credit report data,⁵⁸ the average, median, and 75th percentile credit utilizations are 71%, 82%, and 99%, respectively. As pointed out above, a significant portion of the sample might not have alternate sources of funds⁵⁹ irrespective of whether limited attention or ostrich behavior is at play.

As such, even though our data suggest that more than 50% of NSF fees are incurred over transactions involving less than \$50 (meaning a high cost for small monetary value transactions), we acknowledge that, for some, using overdraft fees is a way to finance their needs. Even though, given the characteristics of our sample, it is challenging to decompose the different theoretical explanations cleanly, we do find evidence for limited attention, and we are able to influence financial decisions for a portion of the experiment population.

6.5. Mixing and Personalizing the Information Design

One possible advantage of the algorithm is that we can test whether there is an intervention policy that matches the users' financial context with intervention type (e.g., simplified avoid or base) to generate an overperforming strategy. Applying such algorithmic analysis might also reveal additional financial characteristics that are relevant to an intervention plan based on different framing. We explore a counterfactual approach utilizing contextual bandit modeling. Contextual bandit modeling is a form of reinforcement learning in which a learner (a system agent) chooses an action, in our case, one type of intervention, based on the user's context. The context is

formed in our use case with similar conceptualized financial attributes that were used for the ODP model. The learner's goal is to find a policy that maps the intervention type to context to maximize rewards, which is reducing overdraft fees. This approach can tell us whether assigning different interventions over time increases the effectiveness of the reminders and whether such a system can find financial characteristics that predict the effectiveness of a particular intervention. We compare the results of the system policy to sending just the simplified avoid alternative, which was the best performing one in our experiment. To implement this approach, we use a premade framework Vowpal Wabbit (e.g., Bietti et al. 2018).⁶⁰ We split our intervention set into train and test sets, in which we train the learner on the train set and evaluate the policy on a test set. Based on our set of three intervention alternatives, we didn't find a system policy powered by the learner that consistently outperforms the simplified negative framing reminder warning.⁶¹

7. Robustness

7.1. Propensity Score Matching Analysis

Although the sample population was randomly distributed across the treatments, groups may still have had different characteristics. For example, first timers may have been categorically different from return ODP users, who had received reminders prior to the experiment. To confirm that there were no characteristics that systematically impacted the main results, we conduct a propensity score matching analysis, which matches between the treatment groups on the number of NSF fees paid and between the first time and return users with parallel yearly incomes.⁶² The results are consistent with our main results: sending a simplified reminder significantly reduces the number of overdraft fees compared with the Base treatment, and the significant differences between the Avoid and the Save framings hold as well. These results support the randomization and suggest that the first timers versus return users allocation did not change the results across the treatment groups.

7.2. Endogeneity of Treatments

We also tested whether the effect of the different versions of the reminder was contingent on the number of messages sent to each participant as decided by the algorithm rather than on their structure or framing. It is noteworthy, in this context, that the group means (and standard deviations) for the number of reminders sent during the experiment were as follows: 3.25 ($SD = 2.75$), 3.15 ($SD = 2.68$), and 3.16 ($SD = 2.71$) for the Base, Simplified Avoid, and Simplified Save group. These statistics mitigate such concern since the largest number of reminders were not sent the most effective message, that is, the Simplified Avoid group.

7.3. Household Income Groups Split

As mentioned above, we could not distinguish between individual and household users. Whereas we analyze the results on the basis of individual income cutoffs; for robustness, we also analyze the results using household income cutoffs⁶³ to see if the effects on the groups remain. We find the same trends but with higher thresholds at every income level. For the immediate response one week after the first reminder, we were able to reduce the number of NSF fees significantly above the 10th income percentile. For the total experiment duration, we significantly reduced the number of NSF fees paid among the 50th percentile and above.

7.4. Participants with a Large Number of NSF's Impacted by the Messages

Participants with a higher total annual income pay more NSF fees, and arguably, the differences observed in the influence of reminders on different income groups depends on the proportion of participants with a prior tendency to pay NSF fees. In other words, the differences observed across the income groups in the impact of the reminders are not due to a particular kind of treatment but occur because income groups differ as to their members' tendency to pay NSF fees in the first place. To address this, we censor our data to create closer NSF fee distributions among the income groups, and we limit the number of weekly fees to seven and five, consecutively. Under both limits, the distribution of NSF fees across the different quantiles is practically the same. The results appear to hold with both censored data analyses: the influence of the reminders is significantly stronger on participants with incomes above the 25th percentile.

8. Conclusions

We investigate whether it is possible to reduce the overdraft fees that individuals are charged by using an AI algorithm to send recurring, as-needed notifications. We conducted a field experiment with 39,607 unique users of Mint, a large web-based personal financial management application in the United States and Canada.

Our goal was to test whether a technology-enhanced, recurring, as-needed reminder mechanism can reduce the limited attention behavioral patterns that affect individuals' financial situations. We conducted two main manipulations: simplification and negative versus positive framing. We find that sending reminders as needed is effective in mitigating limited attention, and that simplifying the message can substantially increase this impact. Our results also reveal that negative framing reduced overdraft fees more effectively than positive framing.

We also show that simplifying the costs of the information-gathering process and sending reminders as needed does not guarantee that individuals will act

upon this information. Participants in our experiment who emerged as the most responsive to the reminders were those with medium to high income levels. For this population, reminders—especially when phrased in simple terms—lowered the cost of gathering information sufficiently to trigger behavior change. Yet we could not obtain the same positive effect for the low-income population. Our findings highlight the need to optimize the human–algo reminding systems, understanding that this solution is beneficial but not always sufficient to overcome financial hardship. We leave it to future research to investigate other mechanisms of personalized AI solutions that take into account not only financial characteristics but also personal characteristics and financial constraints. We likewise leave for future research how our results in this experiment evolve over a longer time period.

Acknowledgments

The authors have benefited from comments by Anat Bracha, Jeremy Burke, Jacob Faber, Ori Heffetz, Avner Kalay, Brian T. Melzer, Olivia Mitchell, Lev Muchnik, Paolina Medina, Michaela Pagel, Eyal Winter, Yishay Yafeh, Stephen Zeldes, and Asaf Zussman as well as the participants at the 2019 AFE University of Chicago; 2019 Exeter Behavioral Prize Workshop, Exeter, United Kingdom; the Intuit AI Summit Conference, 2019; the 2020 FDIC Consumer Research Symposium; 2021 Experimental Finance Society Conference; 2023 Asia ESA meeting, Seoul; Graz University (online, Austria); Hebrew University of Jerusalem (finance and behavioral economics); Kings' College (United Kingdom); and the University of California, Santa Cruz (UCSC)-Bar-Ilan University (BIU)-Ben-Gurion University of the Negev (BGU) seminars.

Endnotes

¹ Examples of this growing literature include Babar et al. (2023) and Calzolari and Nardotto (2017) for wellness and gym interventions, Beketov et al. (2018) for financial robo-advisors, Briganti and Le Moine (2020) and Ghose et al. (2021) for AI and systems in health, Ghose et al. (2019) for commerce and contextual interventions, Iyer (2021) for AI in the transportation space, Kretzer and Maedche (2018) for intervention evaluation in enterprise, Thirumalai and Sinha (2013) and Wang et al. (2022) for e-commerce and personalizing interventions, and Yang et al. (2020) for intervention evaluation and the news space.

² Examples are AI-based trading and investment strategies, robo advisors for asset management, AI solutions for underwriting insurance services, among others.

³ NSF and overdraft fees are commonly aligned and usually range between \$24 and \$35. In most of our inquiry, no distinction is made between the payment of overdraft fees and NSF fees, and these two terms are used interchangeably. Nevertheless, for robustness, using textual analysis, we also differentiate between the two kinds. In our sample and during the experiment, of those who paid fees, 74% paid overdrafts, 66% paid NSFs, and about 40% paid both types.

⁴ Banking systems around the world offer different services for dealing with their customers' overdrafts. In the United States, a common service is overdraft protection, whereby the bank covers certain transactions that trigger an overdraft, mostly in return for

what is termed an "overdraft fee." An NSF fee is imposed when the bank declines the transaction. These two fees apply irrespective of whether the customer is registered for such a protection service. In addition to the overdraft or NSF fee, banks may charge a daily fee or interest for the outstanding negative balance.

⁵ According to the Consumer Financial Protection Bureau (2017), in the United States, overdraft fees amounted to \$17 billion. Several private institutions estimate that amount to be higher: in 2017, consumers paid \$34.3 billion and, in 2016, \$33.3 billion (Moebs Services and Products 2017). According to Melzer and Morgan (2015), in 2007, consumers in the United States paid \$23 billion.

⁶ According to the Consumer Financial Protection Bureau (2017), overdraft fees constituted around two thirds of all retail bank fees.

⁷ Inattention is theoretically investigated with regard to economic decisions in various contexts, for example, Chetty et al. (2009), DellaVigna (2009), Gabaix et al. (2006), Hirshleifer and Teoh (2003), Karlan et al. (2016), and Sims (2010). A more comprehensive review of the potential explanations is found in Section 3.

⁸ The "ostrich effect" bias is initially coined by Galai and Sade (2006) and later extended by others. For examples of information-dependent utility models, see Kőszegi and Rabin (2009) and Andries and Haddad (2019), among others. We provide a more elaborate review of the literature in Section 3.

⁹ See further elaboration about the algorithmic solution in the "setting" section. Also, see the issued patent (Mintz et al. 2021).

¹⁰ Liquidity levels refer to how much money a user has on hand in checking/saving accounts.

¹¹ The results of the investigations of additional time frames and their monetary value are presented in the "results" section (and 3.3% for the duration of the entire experiment of four months).

¹² There was a 9.25% mean reduction after one week and 8.9% after four months in the Simplified Avoid message and 9.44% but only 4.9% for the Simplified Save message over the same time frames. Table 2 provides a high-level summary of the results.

¹³ We follow the Prentice et al. (1981) conditional risk set model for recurring events.

¹⁴ We had data for about 34% of the experiment population.

¹⁵ We didn't have a 100% total credit card utilization rate.

¹⁶ In our setting, due to the potential endogeneity, we can't determine the number of interventions that will be effective.

¹⁷ See the Federal Reserve's opt-in Regulation E.

¹⁸ Gabaix et al. (2006) discusses costly information acquisition, modeling, and comparison between the cognition model and the rational model in the laboratory; DellaVigna (2009) provides models and a survey of empirical evidence in behavioral economics; Chetty et al. (2009) explore limited attention regarding salience and taxation; and Karlan et al. (2016) study theory and evidence regarding the impact of reminders on limited attention and on savings. For a broad review of the literature on inattention, see Caplin (2016) and Gabaix (2017).

¹⁹ Stango and Zinman (2014) present empirical work that is based on administering to panelists questionnaires on overdraft: the objective is to investigate the effect of the survey on overdraft fee payments. Medina (2017) presents empirical work on limited and selective attention in the case of credit card fees. Ashraf et al. (2006) present field experiments on the commitment savings product.

²⁰ Examples of related research include DellaVigna and Linos (2020). Other research examines the impact of single versus multiple interventions (e.g., Wang et al. 2022), the importance of framing concepts and personalizing messages (e.g., Thirumalai and Sinha 2013, Kretzer and Maedche 2018, DellaVigna and Linos 2020, Liu

and KC 2023), and methods for evaluating and analyzing the effectiveness of such systems (e.g., Yang et al. 2020, Wang et al. 2022).

²¹ Also, the user faced at least one overdrawing event in the past.

²² This is based on the algorithm score and up to once a week.

²³ During the experiment, the model was not retrained to avoid selection on participants' entries or with the notification messages sent.

²⁴ See further dissection of the data and representativeness in the "data and experiment design" section.

²⁵ As stated, 81% of participants encountered at least one fee. Additional evidence was found by Grubb et al. (2025), whose UK study compared a similar population with two other groups that have lower rates of overdraft occurrences.

²⁶ Although there is an ongoing debate in the literature over the Mturk sample representativeness and fit for specific tasks, it is a commonly used platform for economics, business, and psychology experiments. We use the Mturk sample for our treatment robustness check. Unlike some other survey methods or interviews, the participants of Mturk are web and internet users. This characteristic is of interest given that, in our primary empirical investigation, we investigate app users. Also, Mint users and the Mturk sample have heterogeneous characteristics, enabling us to obtain answers from a broad range of participants. Last, given that we provided Mturk participants with a straightforward task, we did not anticipate major barriers to performing it. See Goodman et al. (2013) for a detailed discussion about the advantages of using Mturk for tasks such as ours.

²⁷ See Levin et al. (1998) for a literature review. See Massa and Simonov (2005) with respect to assets and financial holdings.

²⁸ The typology distinguishes three types of framing: (a) the original risky choice framework developed by Kahneman and Tversky, (b) attribute framing for relatively good or bad potential outcomes, and (c) goal framing.

²⁹ Medina (2017) explores reminders to address the use case of credit card late payments, Alan et al. (2018) cover bank overdrafts in Turkey, Bursztyn et al. (2018) research incentives in credit card debt repayment in Indonesia, and Grubb et al. (2025) study overdraft SMS messages in the United Kingdom.

³⁰ The project collaboration agreement assured publication rights regardless of the study results.

³¹ The randomization was carried out using a hash function over a random ID number and not over stratified characteristics. Comparing income, credit scores, and credit card utilization levels across treatment groups before the experiment showed no significant difference. ODP is a service provided as part of the product; no opt in was required.

³² No additional preliminary conditions were set for the experiment participants, such as the number of linked accounts, minimum app use duration, demographics, etc. Among the users in the experiment, no attrition was documented.

³³ See Baker (2018) on large account-linked-data representativeness.

³⁴ We calculate the periodic yearly income based on wages, investment interest, dividends, and rentals.

³⁵ Others, such as Crockett et al. (2019) document nonstandard preferences and decision making in a lab experimental setting with complete information. By adding different robustness analyses, field experiment design can be an important method to explore such venues.

³⁶ For robustness, the regression tables also report the results for two weeks and one month after the first alert.

³⁷ In Online Appendix A, there is a table that provides information about additional time frames.

³⁸ In the analysis, we control for yearly income and first timers.

³⁹ This increased open rates by 12.1% and click-to-explore inside the app rate by (via a link in the message) by 78.4%.

⁴⁰ Online Table 11 includes the full regression results.

⁴¹ For a more detailed account of the methodology and comparisons, see Amorim and Cai (2015) and Prentice et al. (1981).

⁴² The effect of yearly income was significantly positive on both the open rate and NSF fees in Models (1), (3), and (4), which means that people with a higher income opened reminders at a greater rate and paid significantly more NSF/overdraft fees. These results were found to be significant in Estimations (6) and (8) as well. *First*—the first timer dummy variable—was found to be negative and significant for Model (1) on the open rate and for Models (3) and (4) on NSF fees. Participants who received the reminder for the first time opened it less, on average, and paid significantly fewer NSF fees. These results were obtained in all the other estimations as evident in Online Tables 7, 8, and 10–12.

⁴³ Based on the U.S. Census Bureau's Current Population Survey (Census CPS) Annual Social and Economic Supplement, data are similar to that of the IPUMS-CPS, University of Minnesota analysis (2019). We use the four personal income quartiles and split the lower and upper quartiles with approximate bottom and top deciles to explore potential notable changes at the tails of the distribution. We excluded 13 participants from this analysis because we do not have their income information. In addition, we censored the sample from outliers of yearly incomes above \$1 million to have the same income distribution across treatments. Notwithstanding that, the full results remain the same when using, for robustness, the entire sample as well. As one might argue that it is debatable if the relevant grouping from the distribution should be based on individual income or household income, we also provide in Section 8 household-level income cuts with a similar analysis. The quality of our results remains. Also, as specified above, and to further address the concern on the completeness of the data, we analyzed the results on customers who linked at least two bank accounts. The results remain.

⁴⁴ Additional descriptives about the income analysis are provided in Online Tables 6 and 7.

⁴⁵ Similar results were observed two weeks and one month after receiving the first reminder.

⁴⁶ It is also consistent with the literature that documents that underprivileged populations often fail to request and, thus, do not receive different benefits to which they are entitled even though this can make a difference to their overall situation. Consistent with our findings, this literature suggests that the simplicity of the information can make a difference. For example, Bhargava and Manoli (2015) claim that take-up is related not only to the frequency but also to the simplicity with which information is provided.

⁴⁷ This is based on the Ascent report: Average Credit Score in America. Data current as of 2019.

⁴⁸ The highest credit score group (800–850) did not show a reduction in NSF, but we observed fewer than 100 users in the control group for this analysis. Also, the fee payment itself may serve as a mechanism to reduce potential inattention to future events, which could be why we observe no reduction effect.

⁴⁹ Asking for a waiver is not considered to be avoiding getting into the negative zone. Also, this option was not suggested in the base intervention and was not used in the fee reduction effect analysis.

⁵⁰ Controlling for credit scores in addition to income reduces the sample size and had no effect on the results.

⁵¹ Overall, about 14,000 users linked one or more credit cards among the experiment population. Our use of linked credit cards and not via credit report total balance differs in the frequency with

which the data gets updated. Also, by looking at the linked account, we have greater certainty that we consistently observe the balance at the end of a particular month.

⁵² See Online Table 14 for detailed results.

⁵³ When controlling for credit scores in addition to income, we reduce significantly the sample size of the analysis. We do find a marginal significant influence of the reminder on money inflows for a one-month period, but we couldn't find a consistent effect across multiple months.

⁵⁴ Controlling for credit scores in addition to income reduces the sample size and had no effect on the results.

⁵⁵ Online Table 13 shows the full regression results.

⁵⁶ Users are assigned the value of one if they opened at least one reminder in the course of the study and zero otherwise. The rate is calculated by dividing the sum of all participants in a group by the group's total number.

⁵⁷ Regression analysis is provided in Online Table 12.

⁵⁸ We have credit report data for about 21,000 of the total experimental population.

⁵⁹ During the experiment, about 38% (33%) of the sample, on average, had more than \$100 in observable liquid cash at the end of each month (and with a credit card utilization rate that is lower than 90%). By expanding the threshold, only 26% (21%) of the users had more than \$500 in available cash (with a credit card utilization rate lower than 75%). About 18.5% (13%) had more than \$1,000, on average, at the end of each month (and a utilization rate below 50%).

⁶⁰ Within Vowpal Wabbit, we used two algorithmic optimization approaches: E-Greedy and Bagging.

⁶¹ For a detailed explanation of the contextual bandits approach we used, see Agarwal et al. (2012).

⁶² We use logit for the confounders and nearest neighbors for the propensity matching.

⁶³ Our analyses are based on Census CPS 2018 data. The cutoffs were as follows: \$0–\$14,600, \$14,600–\$30,700, \$30,700–\$63,000, \$63,000–\$114,000, \$114,000–\$185,000, and \$185,000 and above.

References

- Agarwal A, Dudík M, Kale S, Langford J, Schapire R (2012) Contextual bandit learning with predictable rewards. Lawrence N, Girolami M, eds. *Proc. 15th Internat. Conf. Artificial Intelligence Statist.* (PMLR, New York), 19–26.
- Alan S, Cemalcilar M, Karlan D, Zinman J (2018) Unshrouding: Evidence from bank overdrafts in Turkey. *J. Finance* 73(2):481–522.
- Allen L, Shan Y, Tang Y, Yildirim A (2021) Cutting operational costs by integrating fintech into traditional banking firms. Preprint, submitted January 8, <https://dx.doi.org/10.2139/ssrn.3703840>.
- Amorim LD, Cai J (2015) Modelling recurrent events: A tutorial for analysis in epidemiology. *Internat. J. Epidemiology* 44(1):324–333.
- Andries M, Haddad V (2019) Information aversion. *J. Political Econom.* 128(5):1901–1939.
- Ashraf N, Karlan DS, Yin W (2006) Household decision making and savings impacts: Further evidence from a commitment savings product in the Philippines. Preprint, submitted June 30, <https://dx.doi.org/10.2139/ssrn.912771>.
- Babar Y, Curley S, Ke Z, Liu D, Sheffler Z (2023) The effects of digitally delivered nudges in a corporate wellness program. *J. Assoc. Inform. Systems* 24(1):136–160.
- Baker SR (2018) Debt and the response to household income shocks: Validation and application of linked financial account data. *J. Political Econom.* 126(4):1504–1557.
- Beketov M, Lehmann K, Wittke M (2018) Robo advisors: Quantitative methods inside the robots. *J. Asset Management* 19(6):363–370.
- Benartzi S, Lehrer J (2015) *The Smarter Screen: What Your Business Can Learn from the Way Consumers Think Online* (Hachette UK, London).
- Bietti A, Agarwal A, Langford J (2018) A contextual bandit bake-off. Preprint, submitted February 12, <https://arxiv.org/abs/1802.04064>.
- Bhargava S, Manoli D (2015) Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment. *Amer. Econom. Rev.* 105(11):3489–3529.
- Block LG, Keller PA (1995) When to accentuate the negative: The effects of perceived efficacy and message framing on intentions to perform a health-related behavior. *J. Marketing Res.* 32(2):192–203.
- Bossaerts P, Yadav N, Murawski C (2019) Uncertainty and computational complexity. *Philos. Trans. Roy. Soc. B Biol. Sci.* 374(1766):20180138.
- Bracha A, Meier S (2015) Nudging credit scores in the field: The effect of text reminders on creditworthiness in the United States. Preprint, submitted October 19, <https://dx.doi.org/10.2139/ssrn.2675695>.
- Briganti G, Le Moine O (2020) Artificial intelligence in medicine: Today and tomorrow. *Frontiers Medicine* 7:27.
- Bu D, Hanspal T, Liao Y, Liu Y (2020) Cultivating self-control in fintech: Evidence from a field experiment on online consumer borrowing. Preprint, submitted October 24, 2019, <https://dx.doi.org/10.2139/ssrn.3469571>.
- Bursztyn L, Fiorin S, Gottlieb D, Kanz M (2018) Moral incentives in credit card debt repayment: Evidence from a field experiment. *J. Political Econom.* 127(4):1641–1683.
- Calzolari G, Nardotto M (2017) Effective reminders. *Management Sci.* 63(9):2915–2932.
- Caplin A (2016) Measuring and modeling attention. *Annual Rev. Econom.* 8(1):379–403.
- Chetty R, Looney A, Kroft K (2009) Salience and taxation: Theory and evidence. *Amer. Econom. Rev.* 99(4):1145–1177.
- Consumer Financial Protection Bureau (2017) A closer look: Overdraft and the impact of opting-in. https://files.consumerfinance.gov/f/documents/201701_cfpb_Overdraft-and-Impact-of-Opting-In.pdf.
- Cowan N, Johnson TD, Scott Sauls J (2005) Capacity limits in list item recognition: Evidence from proactive interference. *Memory* 13(3–4):293–299.
- Crockett S, Duffy J, Izhakian Y (2019) An experimental test of the Lucas asset pricing model. *Rev. Econom. Stud.* 86(2):627–667.
- D'Acunto F, Prabhala N, Rossi AG (2019) The promises and pitfalls of robo-advising. *Rev. Futures Markets* 32(5):1983–2020.
- DellaVigna S (2009) Psychology and economics: Evidence from the field. *J. Econom. Literature* 47(2):315–372.
- DellaVigna S, Linos E (2020) RCTs to scale: Comprehensive evidence from two nudge units. NBER Working Paper No. 27594, National Bureau of Economic Research, Cambridge, MA.
- Fisch JE, Turner JA, Center PP (2017) Robo advisers vs. humans: Which make the better financial advisers? Preprint, submitted March 17, <https://doi.org/10.2139/ssrn.2932189>.
- Franco JP, Yadav N, Bossaerts P, Murawski C (2018) Where the really hard choices are: A general framework to quantify decision difficulty. *Animal Behav. Cognition* 5(3):223–242.
- Gabaix X (2017) *Behavioral Inattention* (National Bureau of Economic Research, Cambridge, MA).
- Gabaix X, Laibson D, Moloche G, Weinberg S (2006) Costly information acquisition: Experimental analysis of a boundedly rational model. *Amer. Econom. Rev.* 96(4):1043–1068.
- Galai D, Sade O (2006) The “ostrich effect” and the relationship between the liquidity and the yields of financial assets. *J. Bus.* 79(5):2741–2759.
- Ghose A, Guo X, Li B, Dang Y (2021) Empowering patients using smart mobile health platforms: Evidence from a randomized field experiment. Preprint, submitted February 10, <https://arxiv.org/abs/2102.05506>.

- Ghose A, Kwon HE, Lee D, Oh W (2019) Seizing the commuting moment: Contextual targeting based on mobile transportation apps. *Inform. Systems Res.* 30(1):154–174.
- Goodman JK, Cryder CE, Cheema A (2013) Data collection in a flat world: The strengths and weaknesses of Mechanical Turk samples. *J. Behav. Decision Making* 26(3):213–224.
- Grubb MD, Kelly D, Nieboer J, Osborne M, Shaw J (2025) Sending out an SMS: Automatic enrollment experiments for overdraft alerts. *J. Finance* 80(1):467–514.
- Haran Rosen M, Sade O (2022) Investigating the introduction of a regulatory fintech advancement designed to reduce limited attention regarding inactive saving accounts—Data, survey, and field experiment *AEA Papers Proc.* 112:370–375.
- Hirshleifer D, Teoh SH (2003) Limited attention, information disclosure, and financial reporting. *J. Accounting Econom.* 36(1):337–386.
- Iyer LS (2021) AI enabled applications towards intelligent transportation. *Transportation Engrg.* 5:100083.
- Karlan D, McConnell M, Mullainathan S, Zinman J (2016) Getting to the top of mind: How reminders increase saving. *Management Sci.* 62(12):3393–3411.
- Kőszegi B, Rabin M (2009) Reference-dependent consumption plans. *Amer. Econom. Rev.* 99(3):909–936.
- Kretzer M, Maedche A (2018) Designing social nudges for enterprise recommendation agents: An investigation in the business intelligence systems context. *J. Assoc. Inform. Systems* 19(12).
- L'Esperance M (2023) Nudging credit union members to check their credit: Evidence from a field experiment. *J. Behav. Experiment. Finance* 37:100775.
- Levin IP, Gaeth GJ (1988) How consumers are affected by the framing of attribute information before and after consuming the product. *J. Consumer Res.* 15(3):374–378.
- Levin IP, Schneider SL, Gaeth GJ (1998) All frames are not created equal: A typology and critical analysis of framing effects. *Organ. Behav. Human Decision Processes* 76(2):149–188.
- Lindgaard G, Fernandes G, Dudek C, Brown J (2006) Attention web designers: You have 50 milliseconds to make a good first impression! *Behav. Inform. Tech.* 25(2):115–126.
- Liu J, KC D (2023) Nudging patient choice: Reducing no-shows using waits framing messaging. *Oper. Res.* 71(3):1004–1020.
- Mani A, Mullainathan S, Shafir E, Zhao J (2013) Poverty impedes cognitive function. *Science* 341(6149):976–980.
- Massa M, Simonov A (2005) Behavioral biases and investment. *Rev. Finance* 9(4):483–507.
- Medina PC (2017) Selective attention in consumer finance: Evidence from a randomized intervention in the credit card market. Working paper, Northwestern University, Evanston, IL.
- Melzer BT, Morgan DP (2015) Competition in a consumer loan market: Payday loans and overdraft credit. *J. Financial Intermediation* 24(1):25–44.
- Mintel (2015) Consumers and financial advice – UK – 2015: Consumer market research report. Mintel Group Ltd. <https://www.mintel.com>.
- Mintz I, Hayman L, Mishraky E (2021) Insufficient funds predictor. U.S. Patent No. 11,042,930. U.S. Patent and Trademark Office.
- Moebs Services (2017) Services and products. <http://www.moebs.com/services/nobounce>.
- Ofir M, Wiener Z (2012) Investment in financial structured products from rational and behavioral choice perspectives. Preprint, submitted December 23, <https://dx.doi.org/10.2139/ssrn.2193282>.
- Olafsson A, Pagel M (2017) The ostrich in US: Selective attention to financial accounts, income, spending, and liquidity. NBER Working Paper No. 23945, National Bureau of Economic Research, Cambridge, MA.
- Prentice RL, Williams BJ, Peterson AV (1981) On the regression analysis of multivariate failure time data. *Biometrika* 68(2):373–379.
- Shefrin H (2021) Financial service providers, AI, satisficing, and the human touch in the market for financial nudges and boosts. Preprint, submitted March 18, <https://dx.doi.org/10.2139/ssrn.3779635>.
- Sims CA (2010) Rational inattention and monetary economics. Friedman BM, Woodford M, eds. *Handbook of Monetary Economics*, vol. 3 (Elsevier, Amsterdam), 155–181.
- Sleeman D, Langley P, Mitchell TM (1982) Learning from solution paths: An approach to the credit assignment problem. *AI Magazine* 3(2):2.
- Stango V, Zinman J (2014) Limited and varying consumer attention: Evidence from shocks to the salience of bank overdraft fees. *Rev. Financial Stud.* 27(4):990–1030.
- Thirumalai S, Sinha KK (2013) To personalize or not to personalize online purchase interactions: Implications of self-selection by retailers. *Inform. Systems Res.* 24(3):683–708.
- Tuch AN, Bargas-Avila JA, Opwis K (2010) Symmetry and aesthetics in website design: It's a man's business. *Comput. Human Behav.* 26(6):1831–1837.
- Tversky A, Kahneman D (1981) The framing of decisions and the psychology of choice. *Science* 211(4481):453–458.
- Wang Y, Lu LX, Shi P (2022) Does customer email engagement improve profitability? Evidence from a field experiment in subscription service retailing. *Manufacturing Service Oper. Management* 24(5):2703–2721.
- Yang J, Eckles D, Dhillon P, Aral S (2020) Targeting for long-term outcomes. Preprint, submitted October 29, <https://arxiv.org/abs/2010.15835>.